B.TECH CSE · MAJOR PROJECT

PulseOS

SMART HEALTH MONITORING SYSTEM

Health alerts that learn the patient.

PulseOS is a remote health-monitoring app for patients, doctors and nurses. Its alert engine, **AdaptAlert**, learns each patient's normal, explains every alert in plain words, learns from patient and doctor feedback, and keeps a safety floor so critical readings are never ignored.

~97%	100%	7	2
fewer false alarms than fixed thresholds*	real events caught in the simulation*	stage explainable alert pipeline	languages: English and हिन्दी

* 30-day simulation of 40 virtual patients. Simulated data, not a clinical trial (see Limitations).

Live website	bt39-lemulzeek-nandinisharma04s-projects.vercel.app
Code	github.com/NANDINISHARMA04/bt-39 (branch *main*)
Demo logins	patient@pulseos.app / patient123 · doctor@pulseos.app / doctor123 · nurse@pulseos.app / nurse123
Demo videos	demo-videos/ in the repo: English and Hindi voice-over
Built with	Flutter (web, Android, iOS), Dart, Provider, fl_chart, Supabase (optional), Vercel

1. The project in one minute

The problem. Health monitors alarm whenever a number crosses a fixed line (for example heart rate above 100). At home that happens all the time: a walk, a cup of coffee, a loose strap. Most alarms are false, so patients and staff start ignoring them. This is called **alarm fatigue**, and it is a recognised patient-safety problem (The Joint Commission, Sentinel Event Alert 50).

Our idea. Instead of one threshold for everyone, AdaptAlert learns *this* patient's normal for each activity (rest, light activity, active, cooling down). It alerts only when something is unusual for them, says why in plain language, and learns from feedback: the patient says "I was walking", the doctor marks "false alarm" or "real". Guard rails stop it from learning to ignore real danger.

What we built. A working website (phone and laptop) with three apps in one: patient, doctor and nurse. It also has a live simulated wrist band, a guided demo with voice-over, a Research Lab that runs the experiments for the paper, English/Hindi, a help assistant, and accessibility features for older and disabled people.

2. How AdaptAlert decides (the core of the project)

Every minute the band sends heart rate, SpO_2 (oxygen), temperature and movement. Each window goes through seven stages:




```
P(real) = (1 − w) · P_model + w · P_you w = n / (n + 4) send the alert if P(real) ≥ τ
```

- **P_model**: a decision tree built from clinical reasoning (how many vitals are abnormal, how far from normal, activity, time of day).
- **P_you**: how often alerts of this same pattern were real for this patient, from feedback (Beta posterior). The more answers (n), the more weight it gets.
- **τ (sensitivity)**: personal threshold. A doctor's "false alarm" raises it a little, "real" lowers it more. It is kept between 30% and 75% and can move at most 10 points a week.
- Alerts are **held back**, not deleted: the patient and doctor can still see them, and a worse reading or the safety floor alerts at once.

3. Research gaps it fills

Gap in existing systems	How PulseOS / AdaptAlert addresses it	Where
G1. One threshold for everyone	Personal normal ranges per vital and per activity state	Engine stage C
G2. Systems never learn after deployment	Patient context answers and doctor labels update per-pattern estimates and a personal threshold	Stage F, feedback
G3. Learning can learn to ignore real emergencies	Guarded learning: safety floor that skips the model, bounded and rate-limited threshold, “I feel unwell” never lowers sensitivity	Stage D
G4. Black-box alerts nobody trusts	Plain-language reasons, decision path, counterfactual (“no alert below 84 bpm”), 90% confidence interval	Alert detail page
G5. Bad sensor data causes false alarms	Signal-quality index: unreliable windows never alert or change the normal	Stage A
G6. Personal models start cold	Federated prior: new patients start from label statistics pooled across patients, no raw vitals shared	Research Lab, system E
G7. Results reported without uncertainty	Five systems on identical events, 95% bootstrap confidence intervals, ablation, alarm-fatigue sweep	Research Lab
G8. Apps only in English	Full English and Hindi interface, Hindi health assistant, Hindi voice-over	Whole app
G9. Not usable by older or disabled people	Large text, easy home, flash alerts (hearing loss), spoken alerts (low vision), voice questions, help assistant	Me › Accessibility

G1–G7 are evaluated in the simulation. G8–G9 are built and tested in the app; a user study with older and disabled people would be needed to evaluate them for a paper.

4. Results (simulation)

40 virtual patients × 30 days, identical events for every system: exercise, loose sensor, coffee/stress, night oxygen dips, and 159 real deteriorations (36 at the safety floor). Doctors label 90% of sent alerts.

System	False alarms / patient / day (last week)	Sensitivity	Critical caught	Alerts that were real (PPV)
A. Fixed threshold	3.70	92.5%	100%	3.2%
B. Personal normal only	1.73	100%	100%	7.2%
C. AdaptAlert	**0.11** (95% CI 0.07–0.16)	**100%**	**100%**	**39.6%**
D. Unguarded learning (ablation)	0.11	100%	100%	35.6%
E. AdaptAlert + federated prior	0.13 (week 1: 0.06 vs 0.47)	100%	100%	58.2%

- **~97% fewer false alarms** than fixed thresholds, with every real event caught.
- **~40% of alerts were real**, against ~3% for fixed thresholds.
- **Cold start:** the federated prior cut first-week false alarms from 0.47 to 0.06 per day.
- **Alarm fatigue test:** even if patients dismiss 50% of real alerts, critical events stay at 100% (milder events fall to ~94%).

Limitations (be honest about these in the viva and paper). The numbers come from a simulation with assumed event rates, not real patients. The band in the demo is simulated. The decision tree is hand-built. Next step for a stronger paper: replay real recordings (PhysioNet / MIMIC) and a small volunteer study. Every reference must be checked before citing.

All numbers can be regenerated: `Research Lab` in the app (sliders + “Copy results as Markdown”) or `dart run tool/run_study.dart`. Full write-up and paper outline: `docs/RESEARCH.md`.

5. What the app does

Patient app

- **Today**: live heart rate, oxygen, temperature, activity, signal; “all clear” or alert status
- **Alerts**: every alert with reasons, “What was happening?” feedback, held-back alerts, and “See how AdaptAlert decided”
- **Insights**: trends with the personal-normal band, the live 7-stage pipeline, learned patterns, the study
- **Care**: find doctors (filters), book / reschedule / cancel appointments
- **Me**: medical records upload, insurance & claims, family & caregiver, Medical ID, health profile, settings
- **SOS**: hold to call 112, alert family, share vitals; auto-SOS on critical alerts
- **Help assistant**: answers “where is…/how do I…” in English, Hindi or Hinglish with a button that goes there; mic and read-aloud

Doctor app

- **Today** dashboard: alerts to review, appointments, high-risk ward patients
- **Patients**: search; patient page with vitals, alerts, records, notes; write notes and prescriptions (allergy warning)
- **Schedule**: week view, mark seen with a note, no-show, book follow-ups
- **AdaptAlert review**: mark home alerts real / false → the patient’s model learns

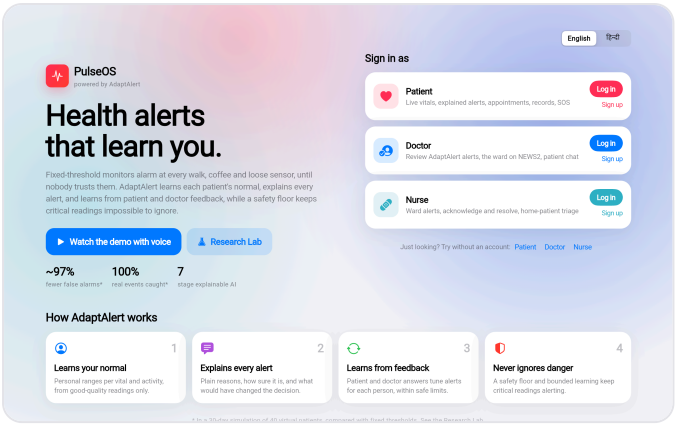
Nurse app

- **My shift**: timed tasks (vitals rounds, medicines, video check), handover notes
- **Ward**: 12 beds sorted by NEWS2 risk; acknowledge and resolve alerts

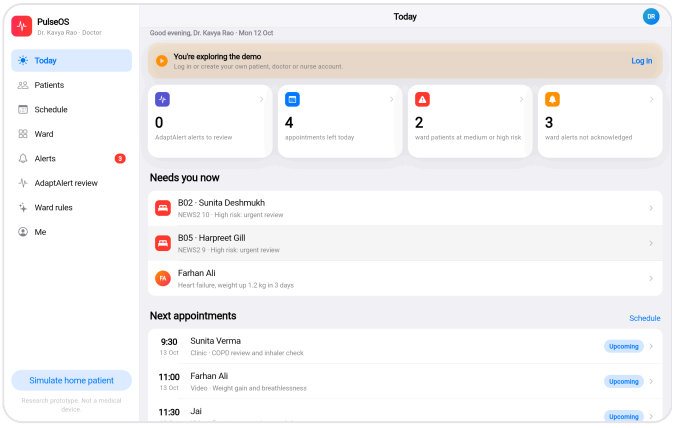
For everyone

- Log in / sign up per role, demo mode, English / हिन्दी
- Accessibility: text size, bold, easy home, flash and spoken alerts, reduce motion
- Guided demo with English or Hindi voice-over, then “Log in now”

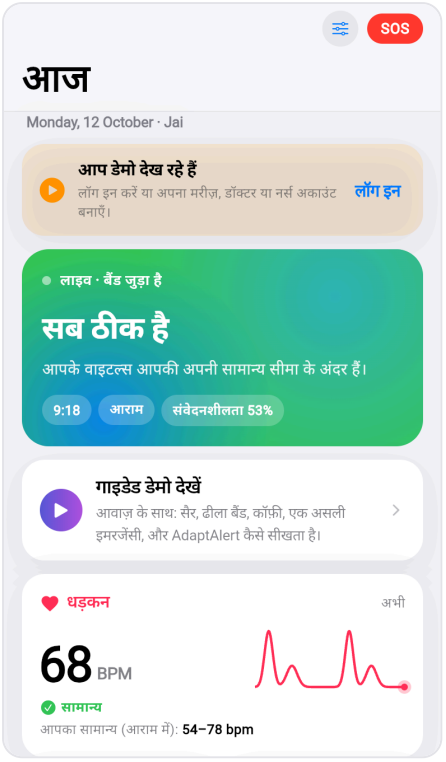
6. Screenshots



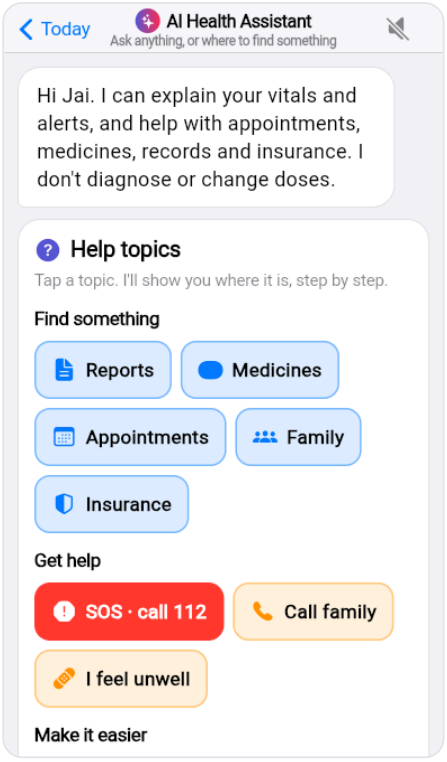
Welcome page (laptop): demo with voice, Research Lab, log in as patient, doctor or nurse.



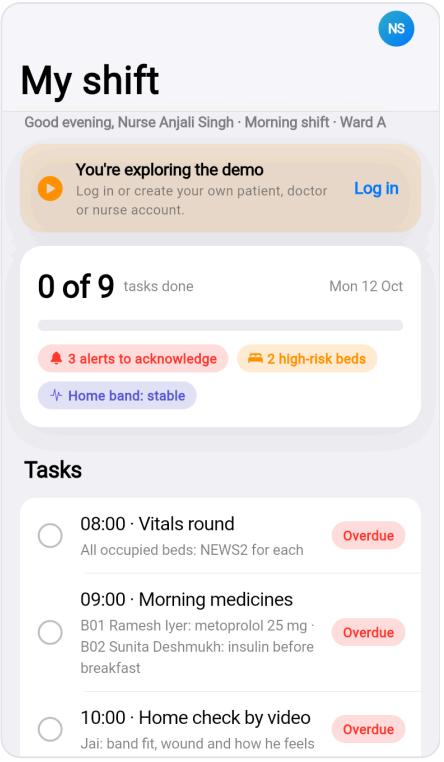
Doctor dashboard: alerts to review, appointments, high-risk patients.



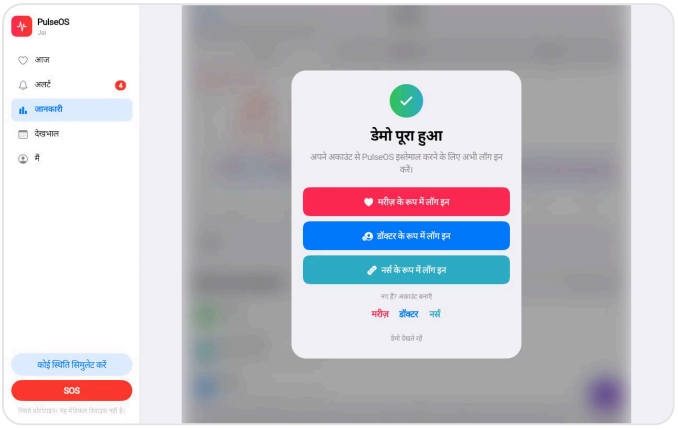
Patient Today in Hindi, with the Help (मदद) button.



Help assistant: topics, mic, step-by-step answers.



Nurse shift: tasks, alerts, handover.



End of the guided demo: “Demo complete — log in now” (Hindi).

Try it yourself

Open the website, press *Watch the demo with voice*, or log in with the demo accounts on page 1. Switch English / हिन्दी at the top of the welcome page.

7. How it is built

Part	Details
App	Flutter 3 (Dart), one code base for web, Android and iOS. iOS-style design, light and dark mode, sidebar on laptop.
State	Provider + ChangeNotifier controllers (MonitorController, PatientStore, ClinicalStore, StaffController, TourController).
AdaptAlert engine	`lib/engine/adapt_alert.dart` pure Dart; band simulator; 30-day study and Research Lab in `lib/engine/study.dart`.
Ward scoring	NEWS2 (Royal College of Physicians, 2017) for the 12-bed demo ward.
Accounts & data	Device accounts with salted SHA-256 password hashes; data in the browser. Optional Supabase (auth, database, realtime) for real multi-device use.
Languages	`tr()` with Hindi maps in `lib/l10n`; a test opens every screen in Hindi and fails on any missing translation.
Voice	Demo voice-over recorded with the open-source Kokoro TTS model (English and Hindi). Help assistant uses the browser's speech recognition and speech.
Quality	100 automated tests (engine, NEWS2, storage, auth, demo, every screen, Hindi, accessibility, assistant).
Hosting	Vercel serves the prebuilt site in `deploy/web`; every push to *main* redeploys. Rebuild with `./tool/build_web.sh`.
Hardware (planned)	ESP32 band: MAX30102 (heart rate, SpO_2), MLX90614 (temperature), MPU6050 (motion) over BLE; or Apple Watch via HealthKit in the iOS app. The engine stays the same.

8. How to demo it (5 minutes)

1. Open the website, choose English or हिन्दी, press **Watch the demo with voice** (sound on). It plays the whole story: walk (no alert), loose band (ignored), coffee (alert + explanation), doctor marks false, same coffee next day (held back), real problem (urgent alert), oxygen 89% (safety floor). It ends with "Log in now".
2. Log in as **doctor**: Today dashboard → Patients → Jai → notes / prescribe → Schedule.
3. Log in as **nurse**: tick a task, write a handover, open the Ward.
4. As **patient**: press **Help** and ask "report kahan hai?"; open **Me › Accessibility** and turn on Easy home and Extra large text.
5. Open the **Research Lab**, run the experiment, show the false-alarm chart and the table.

9. What’s next (and who could take it)

Work item	Why it matters	Suggested owner
ESP32 wrist band (or Apple Watch)	Real readings instead of the simulator; strongest demo for the viva	Hardware / IoT
PhysioNet / MIMIC replay	Validates AdaptAlert on real recordings; biggest boost for publication	ML / data
Small volunteer study (5–10 people, 1 week)	Real before/after alert counts; tests Hindi and accessibility with older users	Research / survey
Supabase backend	Patients and doctors on different devices see the same data	Backend
Paper and report	Use docs/RESEARCH.md: gaps, method, results, outline; verify references	Writing
Concept-drift detection	Notice when a patient’s normal changes (recovery) and tell the doctor	ML

10. References to verify and cite

- Royal College of Physicians. National Early Warning Score (NEWS) 2. London: RCP, 2017.
- The Joint Commission. Sentinel Event Alert 50: Medical device alarm safety in hospitals. 2013.
- Cvach M. Monitor alarm fatigue: an integrative review. Biomedical Instrumentation & Technology, 2012.
- Sendelbach S, Funk M. Alarm fatigue: a patient safety concern. AACN Advanced Critical Care, 2013.
- Drew BJ, et al. Insights into the problem of alarm fatigue with physiologic monitor devices. PLoS ONE, 2014.
- McMahan B, et al. Communication-efficient learning of deep networks from decentralized data. AISTATS, 2017.
- Wachter S, Mittelstadt B, Russell C. Counterfactual explanations without opening the black box. Harvard Journal of Law & Technology, 2018.

PulseOS is a student research prototype, not a medical device. All patients, doctors and clinics in the app are fictional.